SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1:ApplyAI basicssuchasintelligent agents and uninformed search methods to solve problems effectively.*(BL3)*
Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:
I Y.CHRISTINA (1965260124)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand thatanyviolation of academic integrity rules will result in disciplinary action.
Signature of the student:

Christina.y

1. DOCTOR SHIFT ASSIGNMENT

Problem: Assign D1, D2 and D3 to Morning, Afternoon and Night shifts while satisfying all constraints.

Constraints

- D1 cannot work the Night shift.
- D2 must work before D3: Morning < Afternoon < Night. •

Only one doctor is assigned to each shift.

- D3 cannot work the Morning shift.
- Every doctor must be assigned exactly one shift.

Backtracking steps

Step	Assignment	Result
1	D2 = Morning	Valid so far.
2	D3 = Afternoon	D1 would be Night, violating D1≠ Night. Backtrack.
3	D3 = Night	Valid.
4	D1 = Afternoon	All constraints satisfied.

Final valid schedule

Doctor	Shift
D1	Afternoon
D2	Morning
D3	Night

Answer: D2 → Morning, D1 → Afternoon, D3 → Night.

2. ROBOT NAVIGATION USING BFS

Given: Arobotmovesina 5×5grid fromStart(S)to Goal (G). It can move Up, Down, Left and Right. Each move costs 1 and obstacles cannot be crossed.

Important: The supplied page does not show the actual 5×5 grid, obstacle positions, or S/G coordinates. Therefore, an exact path and numerical cost cannot be determined without inventing missing data.

BFS procedure

- Place S in the FIFO queue and mark it visited.
- Remove the front node.
- Generate valid Up, Down, Left and Right neighbours.
- Reject outside-grid, obstacle and already-visited cells.
- Store each new cell's parent and enqueue it.
- Continue until G is reached.
- Follow parent pointers from G to S. Since every move costs 1, the number of moves is the optimal cost.

Queue updates

Stage	Operation	Queue
1	Insert S	[S]
2	Expand S	Add valid neighbours
3	Expand oldest node	Append valid neighbours
4	Repeat	FIFO order maintained
5	Reach G	Stop and reconstruct path

Optimality: BFS is optimal because every move has equal cost. Manhattan distance may be used as a heuristic, but standard BFS does not use it for node ordering.

Final answer: The exact optimal path and cost are undetermined because the grid layout is missing.

3. AUTONOMOUS RESCUE ROBOT

Problem: The robot works in a partially observable 5×5 grid. Some cells are blocked, it senses adjacent cells, each new cell costs 1, entering a risky zone adds 2, and the robot must minimize total cost while reaching survivor G.

A) ANALYSIS OF CONSTRAINTS

Constraint	Effect on decision-making
Movement	Only Up, Down, Left and Right actions are available.
Blocked cells	Blocked cells cannot be selected.
Partial observability	The complete map is unknown initially.
Limited sensing	Only adjacent cells can be observed.
Normal cost	Each ordinary move costs 1.
Risk penalty	A risky entry costs 1 + 2 = 3.
Dynamic knowledge	The map must be updated after sensing.
Goal	Reach G with minimum safe accumulated cost.

B) SOLUTION APPROACH — ONLINE UNIFORM COST SEARCH (UCS)

- Initialize at S with cost 0 and a partial map.
- Sense adjacent cells and classify them as free, blocked or risky.
- Generate valid moves: normal = 1; risky = 3.
- Store candidates in a priority queue ordered by cumulative cost g(n).
- Expand the lowest-cost known state.
- Execute only the next action, then sense again.
- Update the map with newly discovered information.
- If the route becomes invalid, re-run UCS using the updated map.
- Continue until G is reached.

Example: A route with 5 moves, including one risky entry, costs 4×1 + 3 = 7.

C) AI CONCEPTS

Concept Intelligent	Explanation
Agent Rationality	The robot senses, decides and acts to reach the survivor.
Environment Type	It chooses the lowest-cost safe action using available information.
	Partially observable, sequential, discrete and uncertain from the robot's perspective.

Concept	Explanation
Search / Planning	UCS performs cost-based planning; online replanning adapts to new information.

D) JUSTIFICATION

OnlineUniform CostSearch is most suitable because the problem has unequal costs and incomplete information. BFS minimizes the number of moves, but it cannot minimize total cost when risky cells have extra penalties. UCS expands the lowest cumulative-cost path, while online replanning allows the robot to respond when sensors reveal new obstacles or risks.

Final answer: Use an ***Online Uniform Cost Search with dynamic replanning***. It handles limited sensing, blocked cells, unequal costs, risk penalties and minimum-total-cost requirements.